higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

18 August 2025 (X-paper)
09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists of 6 pages, a 2-page formula sheet and a 5-page graph sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled.
5. Use only a black or blue pen.
6. Write neatly and legibly.

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SECTION A: FLOW MEASUREMENT

QUESTION 1

1.1 The flow rate of fluid flow can be expressed in two ways, namely volume flow rate and mass flow rate.

1.1.1 Explain the difference between *volume flow rate* and *mass flow rate*.

1.1.2 Write down the mathematical formulae for volume flow rate and for mass flow rate.

(2 × 2) (4)

1.2 There are two major types of ultrasonic flowmeter measuring principles in use today, the transit-time method and the Doppler shift method.

1.2.1 Briefly describe the construction of an ultrasonic flowmeter. (4)

1.2.2 Explain the principle for the transit-time ultrasonic flowmeter. (6)

[14]

QUESTION 2

Calculate the flow rate of water in a pipeline, based on the following information:

Maximum differential pressure = 2 000 Pa
Inside diameter of orifice plate = 20 cm
Outside diameter of orifice plate = 30 cm
Line pressure = 30 kPa
Water temperature = 30 °C
D + D/2 tapping are used.

2.1 Calculate the area relation (m). (2)

2.2 Determine the discharge coefficient. (1)

2.3 Calculate the velocity approach factor (E). (2)

2.4 What is the value of the expansion factor (ε)? Give a reason for your answer. (2)

2.5 Calculate Reynold s number (R_d). (5)

2.6 Calculate the correction factor (Z). (3)

2.7 Calculate the flow rate of water. (3)

[18]

TOTAL SECTION A: 32

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SECTION B

QUESTION 3: DENSITY, HUMIDITY AND VISCOSITY MEASUREMENT

3.1 Define the following terms:

3.1.1 Density

3.1.2 Humidity

3.1.3 Viscosity

3.1.4 Newtonian fluids

3.1.5 Dew point

(5 × 2) (10)

3.2 The hydrometer operates on the principle that the differential head uses two tubes placed at different levels inside a container, and air is purged through the tubes.

3.2.1 Make a neatly labelled hydrometer sketch that matches the description above. (4)

3.2.2 Explain the principle of operation of the hydrometer described in the above statement. (5)

3.3 Make a neatly labelled sketch of a capacitive hygrometer for solids. (4)

3.4 Explain the operating principle of the falling piston viscometer. (8)

[31]

QUESTION 4: pH AND CONDUCTIVITY MEASUREMENT

4.1 Explain the term *oxidation-reduction reactions.* (2)

4.2 Write down the general form of the equation of oxidation-reduction reactions. (2)

4.3 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (4.3.1–4.3.4) in the ANSWER BOOK.

4.3.1 In a basic solution, the value of a hydroxyl ion concentration is greater than that of a hydrogen ion concentration.

4.3.2 pH 7 indicates a neutral solution.

4.3.3 A pH value of less than seven indicates a basic solution.

4.3.4 A pH value greater than seven indicates an acidic solution.

(4 × 1) (4)

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4.4 The pH of an aqueous solution is expressed as $pH = -\log\left[H_3O^+\right]$

4.4.1 Explain the meaning of the pH equation above. (2)

4.4.2 Calculate the amount of hydrogen ions present in a mixture with a pH of 6. (3)

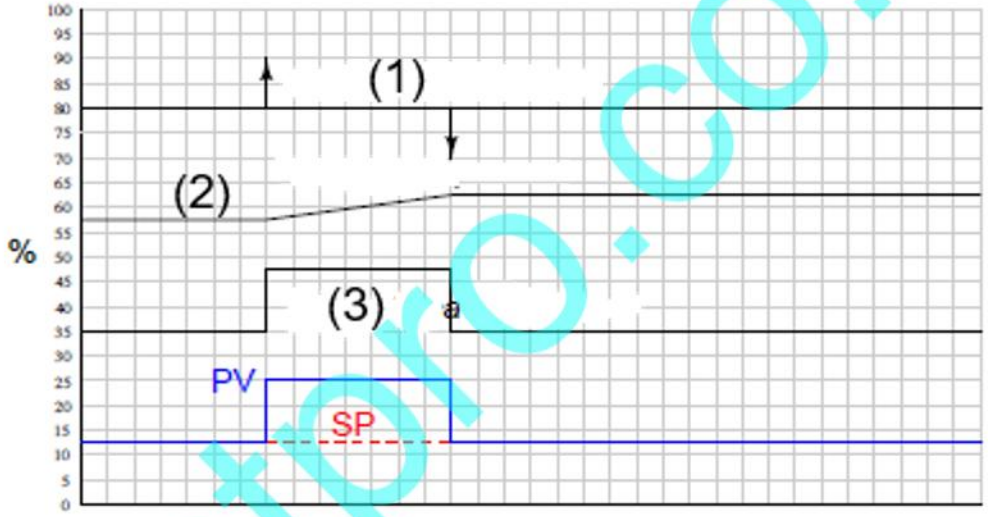
4.4.3 What is the range of a pH scale? (1)

[14]

QUESTION 5: CONTROL SYSTEMS

5.1 Name FOUR basic functions of an automatic control system (4)

5.2 Responses for a momentary step-and-return are shown by the graphs below.



What type of control action is indicated by the response (1), (2) and (3)? (3)

5.3 Explain the functions of the following control actions:

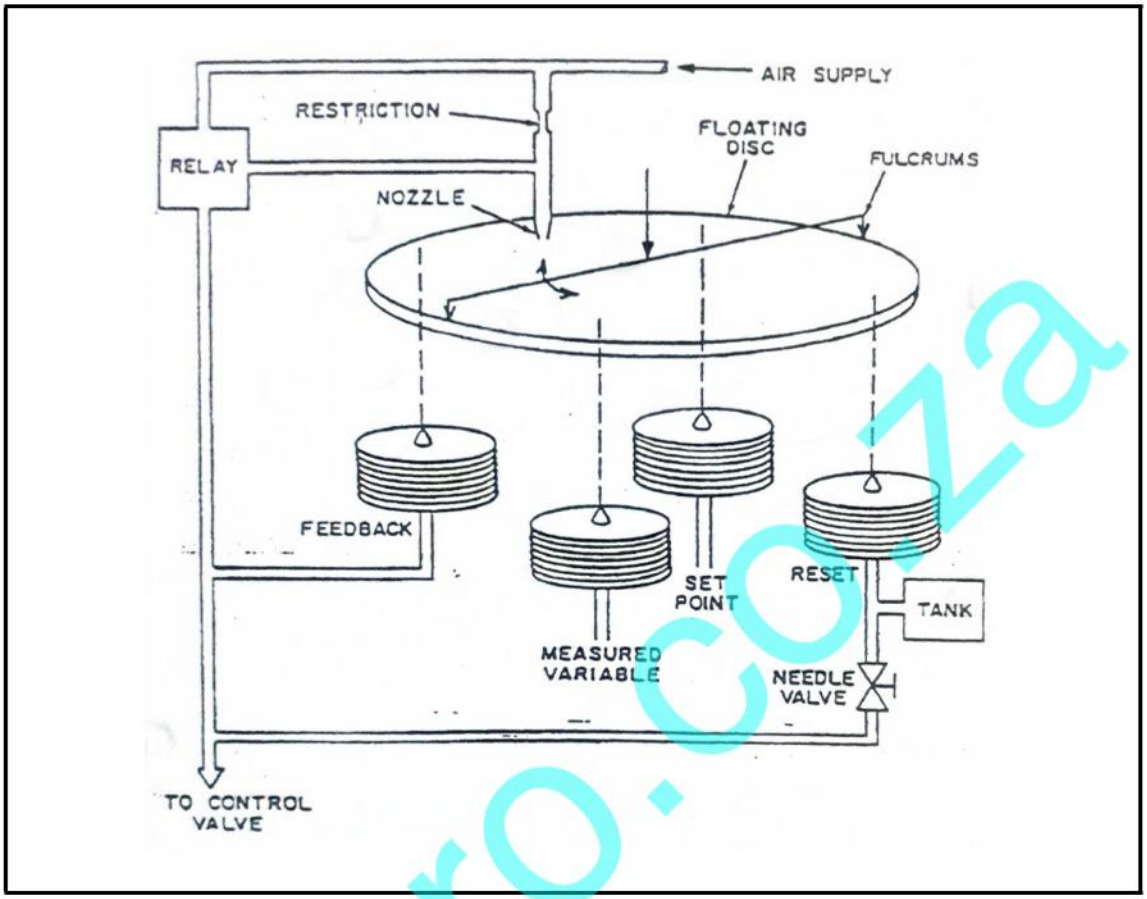
5.3.1 Integral control action

5.3.2 Derivative control action

(2 × 2) (4)

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5.4 The sketch below shows a typical moment balanced control for a pneumatic receiver controller.



5.4.1 Explain the conditions under which the controller is balanced. (3)

5.4.2 What will happen if the measured variable is increased by 10 kPa, the gain setting is 1 and direct? No integral or derivative action is generated. (3)

5.4.3 What will happen if the measured variable is increased by 10 kPa, the gain setting is 1 and indirect? No integral or derivative action is generated. (3)

5.4.4 Calculate the proportional band of the controller if the gain setting is as in QUESTION 5.4.2. (1)

5.5 What is the difference between an *error signal* and *measurable variable signal*? (2)

[23]

TOTAL SECTION B: 68
GRAND TOTAL: 100

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INDUSTRIAL INSTRUMENTS N5

FORMULA SHEET

$W = 359·2\ CZ·Ed^2\sqrt{(h\rho)}$

$Q = 359·2\ CZ·Ed^2\sqrt{(h/\rho)}$

$W = 0·01252\ CZ·Ed^2\sqrt{(h\rho)}$

$Q = 0·01252\ CZ·Ed^2\sqrt{(h/\rho)}$

$Q_g = 2238\ CZEd^2\sqrt{(h/\rho)}$

$m = (d/D)^2$

$R_d = W/15·8\ \mu d$

$R_d = Q·/15·8\ \mu d$

$R_d = 3·54\ W/\mu d$

$R_d = 3·54\ Q·/\mu d$

$R_d = Q_g·/98·6\ \mu d$

$E = 1/\sqrt{(1 - m^2)}$

$$N = \frac{W}{0·01252\ D^2\sqrt{(h\rho)}} = \frac{Q\sqrt{(\rho)}}{0·01\ 252\ D^2\sqrt{(h)}}$$

$mE = N/CZ·$ $\quad$ $CmE = N/Z·$ $\quad$ $mE = CmE/C$

$m = (d/D)^2$ $\quad$ $E = 1/\sqrt{(1 - m^2)}$

$$R_d = \frac{W}{15·8\ \mu D\sqrt{(m)}} = \frac{Q\rho}{15·8\ \mu D\sqrt{(m)}} = \frac{Q_g\rho}{98·6\ \mu D\sqrt{(m)}}$$

$$N = \frac{W}{359·2\ D^2\sqrt{(h)}} = \frac{Q\sqrt{(\rho)}}{359·2\ D^2\sqrt{(h)}} = \frac{Q_g\sqrt{(\rho)}}{2238\ D^2\sqrt{(h)}}$$

$d/D = [(mE)^2/1 + (mE)^2]^{\frac{1}{4}}$ $\quad$ $mE = N/CZ·$

$W = 1890\ Ud^2\sqrt{(\rho P)}$ *for critical flow*

$CmE = N/Z·$ $\quad$ $mE = CmE/C$

$d = [W/1890\ U\sqrt{(\rho P)}]^{\frac{1}{2}}$ *for critical flow*

$$R_d = \frac{3·54\ W}{\mu D\sqrt{(m)}} = \frac{3·54\ Q\rho}{\mu D\sqrt{(m)}}$$ $\quad$ $d/D = [(mE)^2/1 + (mE)^2]^{\frac{1}{4}}$

$W = 1·252\ Ud^2\sqrt{(\rho P)}$ *for critical flow*

$d = [W/1·252\ U\sqrt{(\rho P)}]^{\frac{1}{2}}$ *for critical flow*

1 kPa = 102 mmWD = 102 mmWG

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1 lb/ft³ = 16,0183 kg/m³

Atmospheric pressure = 101,325 kPa

Gravitation acceleration = 9,81 m/s²

For D + D/2 tappings and tappings

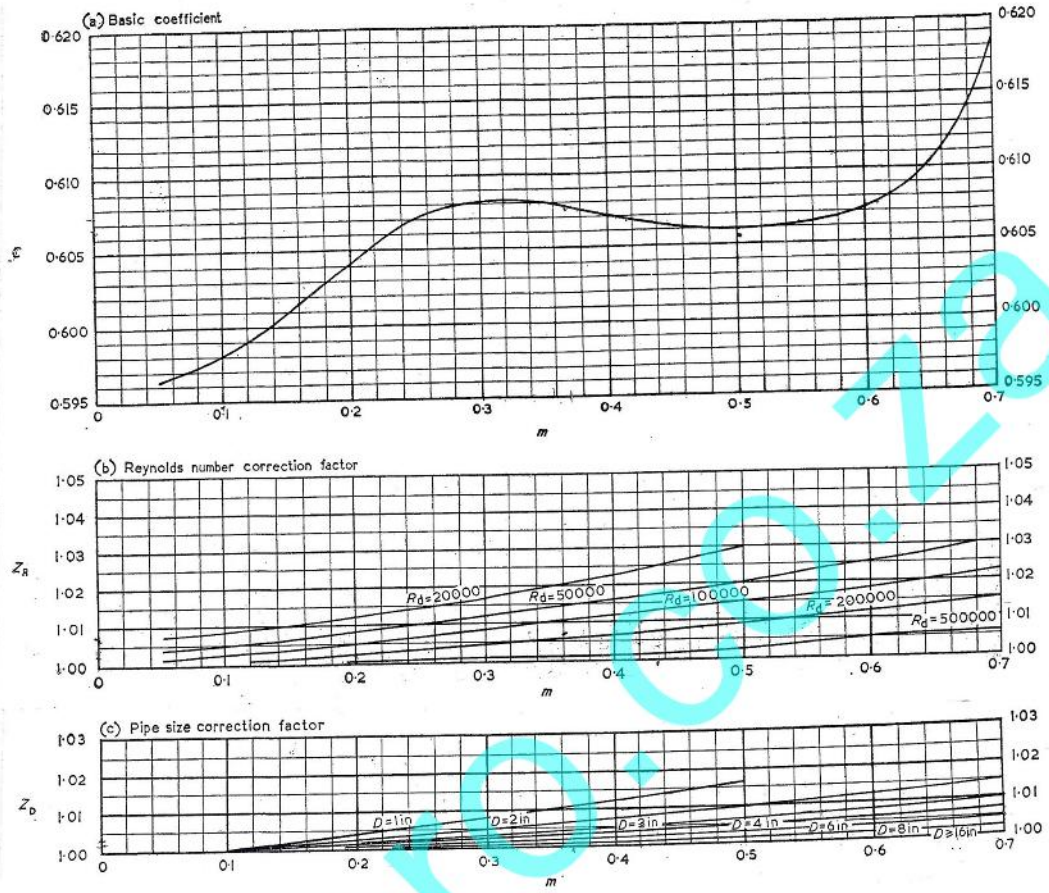
$$\frac{h}{Pa} \times 27,2 = \frac{kPa}{kPa} \times 27,2$$

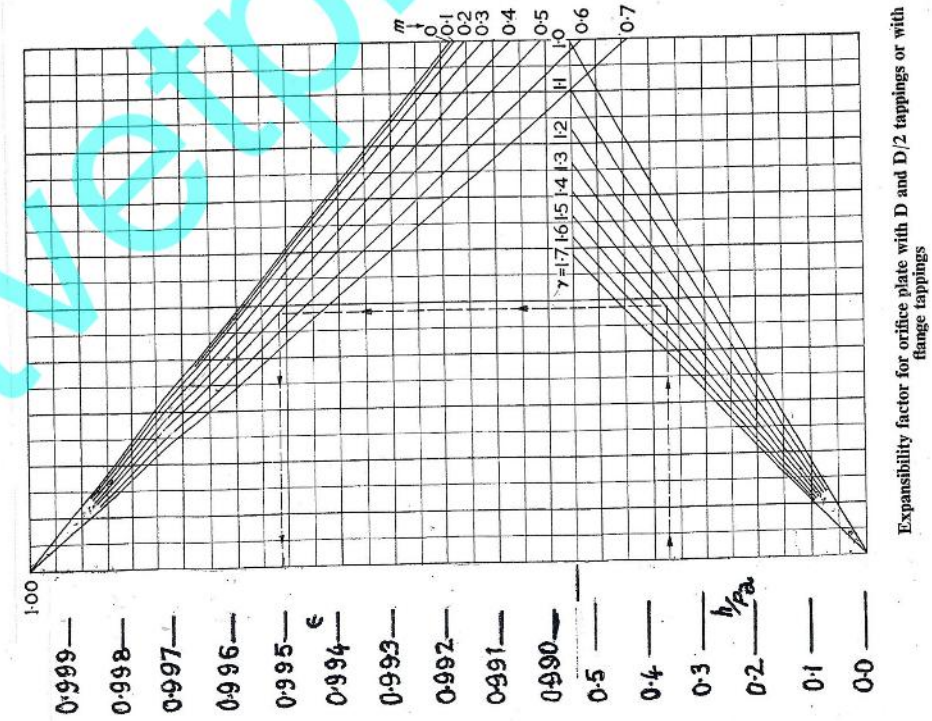
$$Q = \frac{8}{15} \ Tan \frac{\theta}{2} \sqrt{2g.H^5} \qquad Q = \frac{2}{3} \ B \sqrt{2g.H^3}$$

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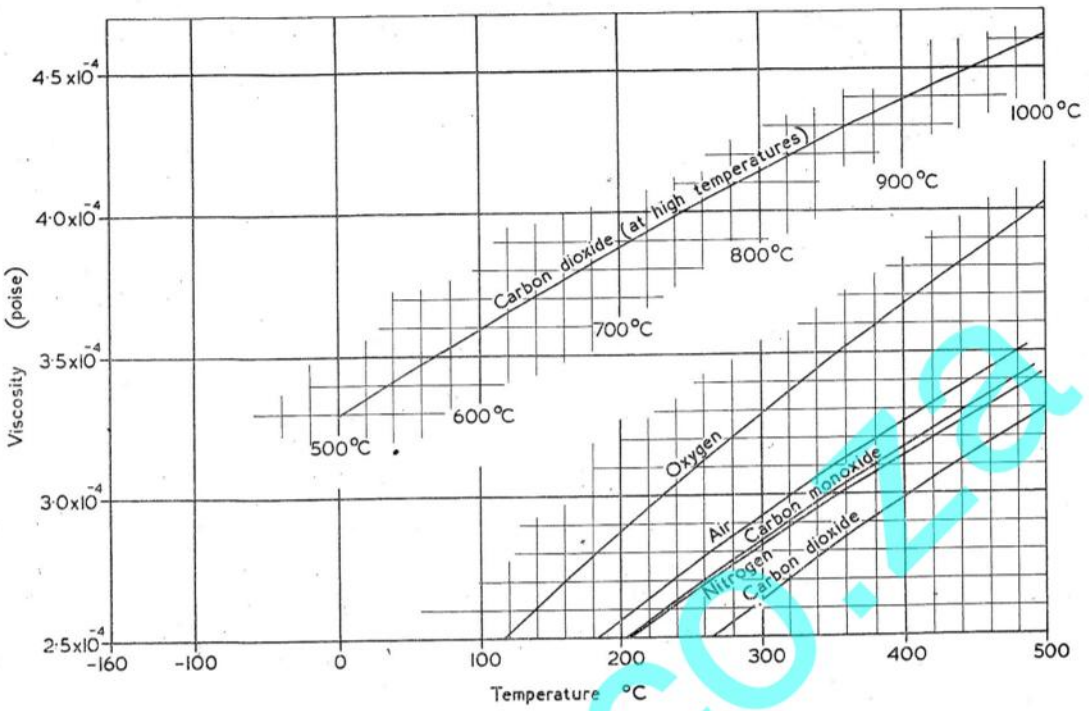
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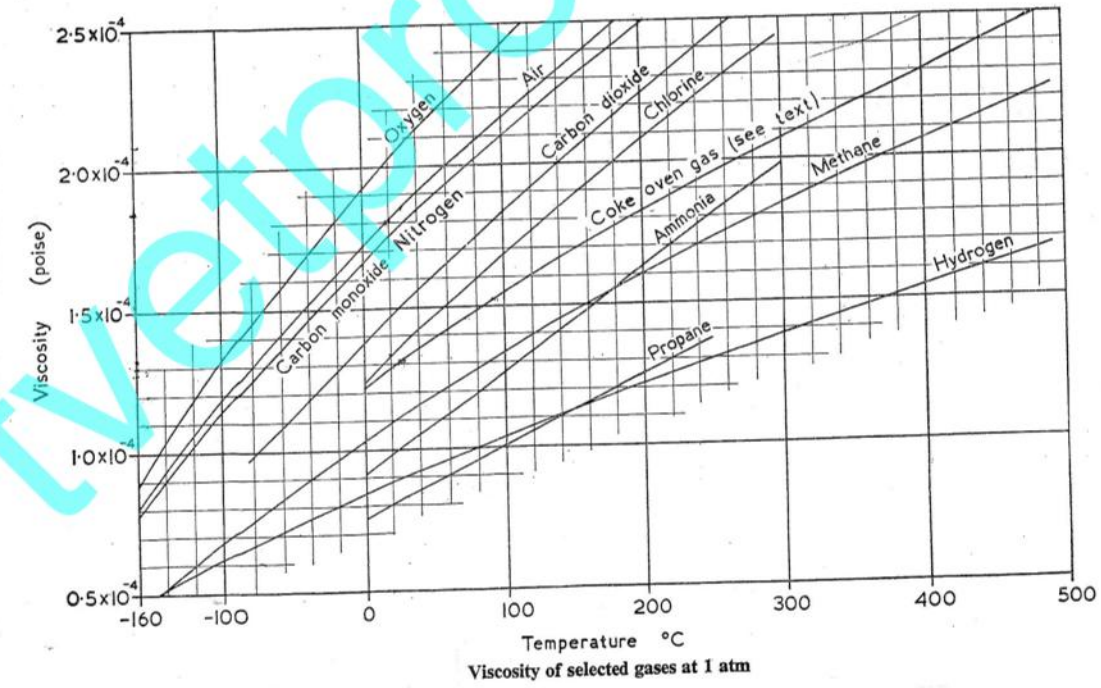
GRAPH SHEET





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Viscosity of selected gases at 1 atm

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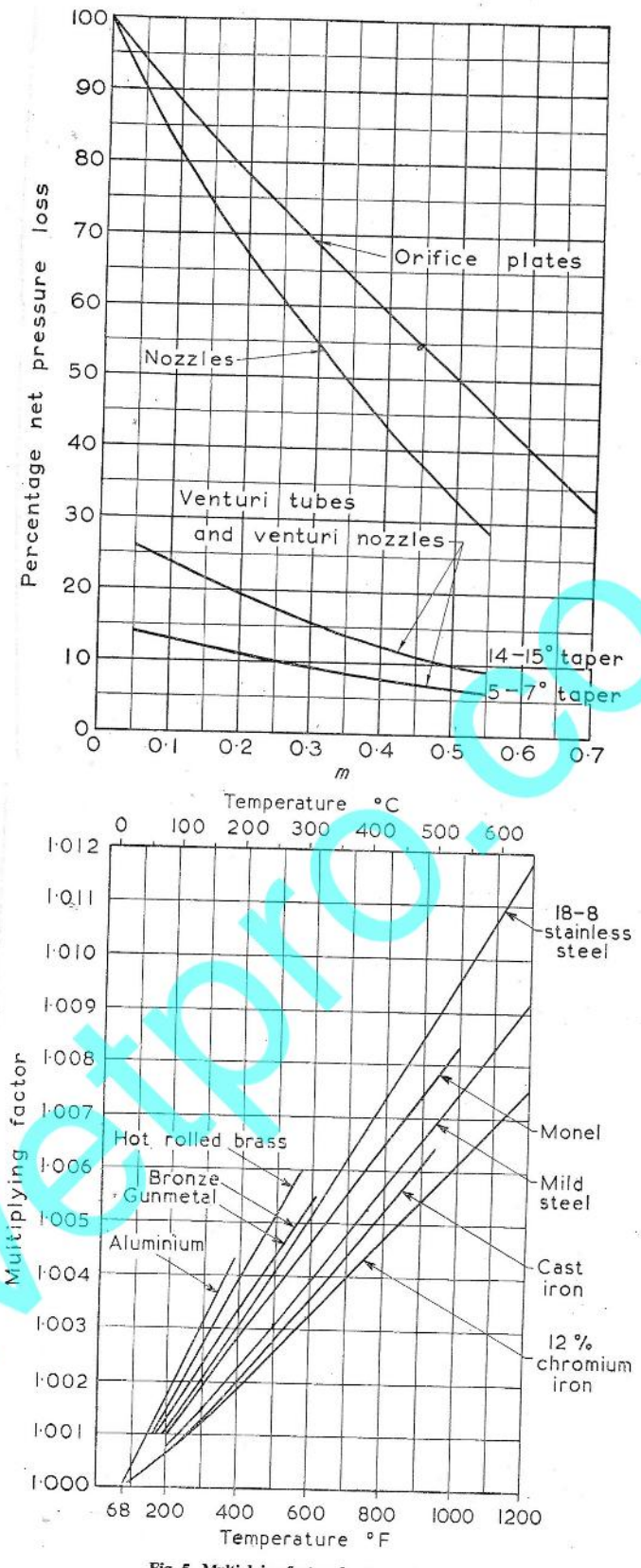
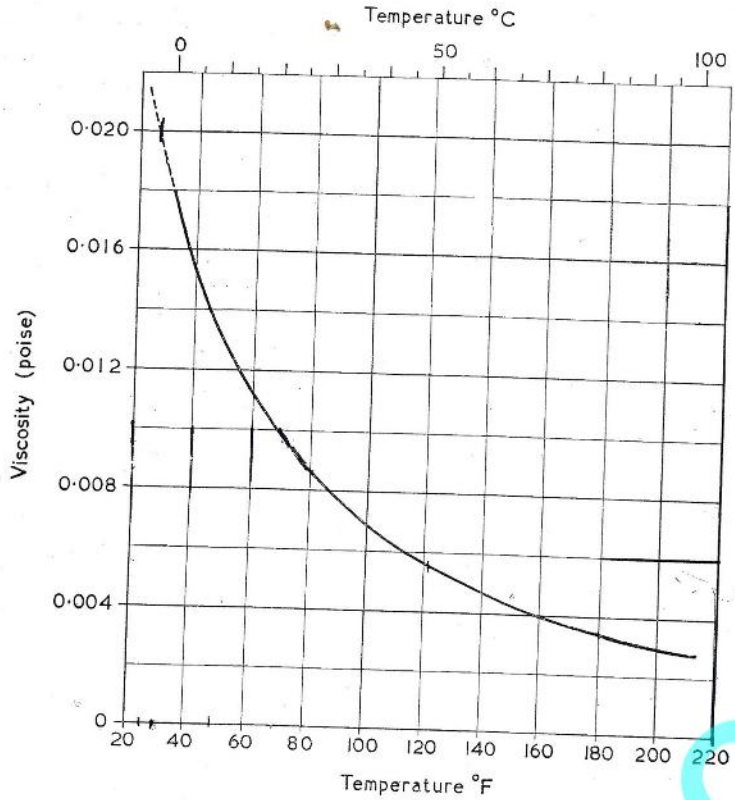


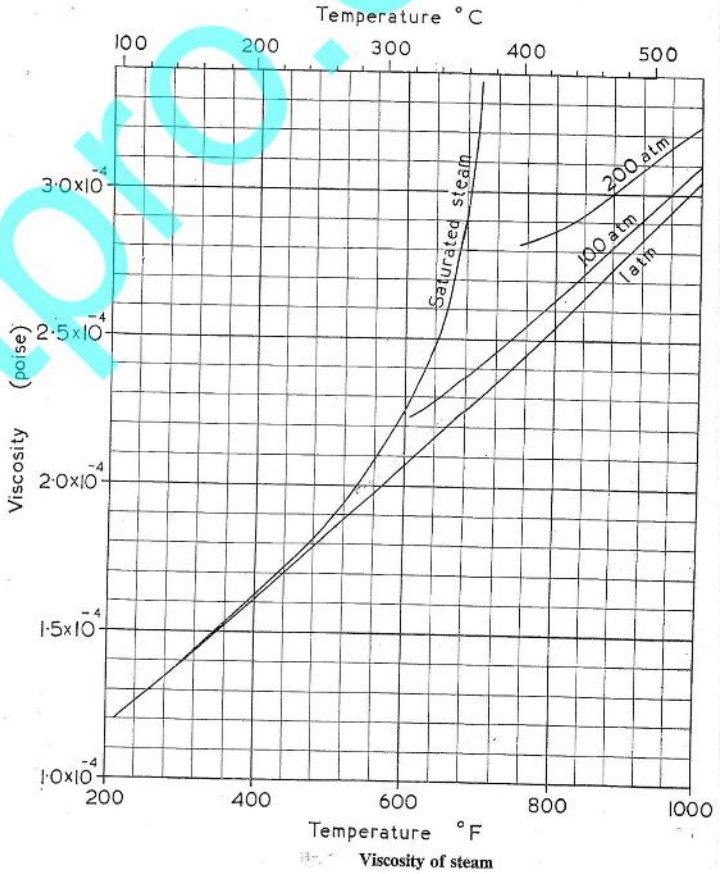
Fig. 5. Multiplying factors for thermal expansion

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Viscosity of water at 1 atm

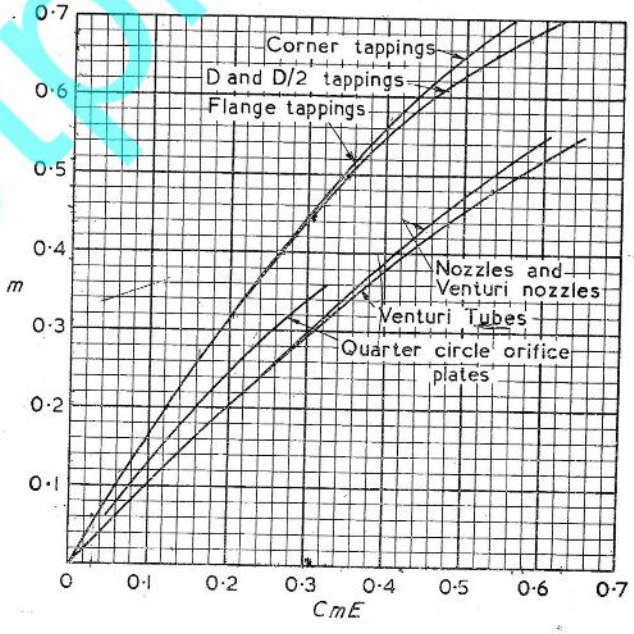


Viscosity of steam

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TABLE 5. DENSITY (IN LB/FT³) OF SATURATED AND SUPERHEATED STEAM*

Absolute pressure (lbf/in²)	Saturated steam	Superheated steam at a total temperature of								
		250°F	300°F	350°F	400°F	450°F	500°F	550°F	600°F	650°F
14	0·03559	0·03351	0·03121	0·02922	0·02749	0·02595	0·02458	0·02335	0·02224	0·02123
15	0·03795	0·03593	0·03347	0·03133	0·02946	0·02781	0·02634	0·02502	0·02383	0·02275
16	0·04032	0·03834	0·03571	0·03342	0·03143	0·02966	0·02810	0·02669	0·02542	0·02427
17	0·04266	0·04080	0·03798	0·03555	0·03342	0·03154	0·02987	0·02837	0·02702	0·02579
18	0·04500	0·04322	0·04023	0·03765	0·03539	0·03339	0·03162	0·03004	0·02860	0·02731
19	0·04735	0·04566	0·04248	0·03976	0·03737	0·03526	0·03340	0·03172	0·03019	0·02883
20	0·04968	0·04810	0·04474	0·04186	0·03934	0·03713	0·03516	0·03339	0·03180	0·03034
22	0·05432	0·05299	0·04926	0·04608	0·04329	0·04085	0·03868	0·03674	0·03498	0·03338
24	0·05893	0·05790	0·05382	0·05033	0·04726	0·04458	0·04221	0·04008	0·03815	0·0364^
26	0·06353	0·06281	0·05838	0·05456	0·05123	0·04833	0·04575	0·04342	0·04136	0·0394(
28	0·06812	0·06775	0·06289	0·05879	0·05522	0·05206	0·04929	0·04679	0·04456	0·0425(
30	0·07273	0·07267	0·06748	0·06305	0·05917	0·05580	0·05283	0·05015	0·04776	0·04556
32	0·07722	—	0·07215	0·06739	0·06321	0·05959	0·05640	0·05353	0·05097	0·04861
34	0·08177	—	0·07675	0·07163	0·06720	0·06337	0·05995	0·05691	0·05417	0·05168
36	0·08621	—	0·08130	0·07587	0·07117	0·06711	0·06349	0·06024	0·05737	0·05473
38	0·09066	—	0·08591	0·08019	0·07519	0·07087	0·06702	0·06361	0·06057	0·05777
40	0·09515	—	0·09058	0·08446	0·07924	0·07468	0·07062	0·06698	0·06378	0·06083
42	0·09970	—	0·09524	0·08881	0·08326	0·07843	0·07413	0·07037	0·06698	0·06390
44	0·1041	—	0·09990	0·09311	0·08726	0·08217	0·07770	0·07375	0·07022	0·06693
46	0·1086	—	0·1045	0·09737	0·09132	0·08598	0·08137	0·07716	0·07342	0·07003
48	0·1130	—	0·1093	0·1017	0·09542	0·08985	0·08489	0·08052	0·07663	0·07305
50	0·1174	—	0·1140	0·1061	0·09950	0·09363	0·08849	0·08389	0·07987	0·07616
52	0·1218	—	0·1187	0·1104	0·1035	0·09737	0·09208	0·08734	0·08313	0·07924
54	0·1262	—	0·1234	0·1148	0·1075	0·1012	0·09560	0·09074	0·08636	0·08230
56	0·1305	—	0·1281	0·1192	0·1116	0·1050	0·09921	0·09416	0·08952	0·08540
58	0·1349	—	0·1329	0·1236	0·1157	0·1089	0·1029	0·09756	0·09276	0·08842
60	0·1393	—	0·1377	0·1280	0·1198	0·1127	0·1065	0·1009	0·09597	0·09149



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